AQA 3.2.2 and 3.2.3 Tests for ions (required practical 4)

Answer in AQA exam style: use the marks as a guide to how much to write and show all working in calculations. Mark your work with the mark scheme at the back. Questions marked as practice questions in the mark scheme were written in AQA style; all others are from AQA past papers.

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19 marks · about 25 minutes

Key information

- CO_3^{2-}: dilute acid gives effervescence; CO_2 turns limewater cloudy
- SO_4^{2-}: add HCl-acidified $BaCl_2(aq)$ — white precipitate; the acid removes CO_3^{2-} first
- Halides with HNO_3-acidified $AgNO_3(aq)$: Cl^- white, Br^- cream, I^- yellow; F^- no precipitate
- Silver halides with $NH_3(aq)$: AgCl dissolves in dilute, AgBr only in concentrated, AgI in neither
- OH^-: red litmus turns blue / universal indicator turns blue-purple

Section A: Short-answer questions

01 Give the ionic equation, including state symbols, for the reaction of silver ions with iodide ions.

[1 mark]

02 Which pair of solutions, when mixed, reacts to form a dark brown solution?

[1 mark]

A $NaF(aq) + Cl_2(aq)$
B $NaCl(aq) + Br_2(aq)$
C $NaBr(aq) + Cl_2(aq)$
D $NaI(aq) + Br_2(aq)$

03 A test for chloride ions in aqueous solution involves adding dilute nitric acid followed by aqueous silver nitrate. What is the reason for adding the nitric acid?

[1 mark]

A To convert $AgNO_3$ into $[Ag(NO_3)_2]^-$
B To decrease the solubility of silver chloride
C To increase the pH of the solution
D To prevent the precipitation of other silver compounds

04 State what is observed when silver nitrate solution is added to sodium fluoride solution.

[1 mark]

05 Describe how dilute sulfuric acid can be used to distinguish between solutions of magnesium nitrate and barium nitrate. State the observation with each solution.

[2 marks]

06 Describe a test to distinguish between solutions of barium chloride and potassium chloride. Give the observation with each.

[2 marks]

Section B: Longer questions

07 Four unlabelled bottles contain solutions of ammonium chloride, sodium carbonate, magnesium sulfate and potassium bromide. A student labels the solutions W, X, Y and Z and does three tests on separate samples of each. The table shows the results.

	Add dilute HNO_3 then $AgNO_3(aq)$	Add acidified $BaCl_2(aq)$	Warm with NaOH(aq); damp red litmus
W	white precipitate	no visible change	litmus turns blue
X	effervescence	effervescence	no visible change
Y	no visible change	white precipitate	white precipitate, litmus stays red
Z	cream precipitate	no visible change	no visible change

07.1 Identify W, X, Y and Z.

[2 marks]

07.2 Give an ionic equation for the reaction that produces the gas in the test on W.

[1 mark]

07.3 Give an ionic equation for the formation of the white precipitate when NaOH(aq) is added to Y.

[1 mark]

07.4 Describe a further test on the precipitate formed from Z that would confirm that it contains bromide ions rather than chloride ions.

[2 marks]

08 A student does a series of reactions with aqueous solutions of some potassium halides (P, Q and R) of equal concentration. Each solution contains a different halide ion (chloride, bromide or iodide).

The student adds 3 drops of bromine water to 3 drops of each solution, and also to 3 drops of water. The table shows the observations.

	Observation when bromine water is added
Solution P	Orange solution
Solution Q	Brown solution
Solution R	Orange solution
Water	Orange solution

08.1 Identify the halide ion present in Q. Give the ionic equation for the reaction that occurs when bromine water is added to Q.

[2 marks]

08.2 In a second experiment the student adds a few drops of aqueous silver nitrate solution to 2 cm^3 of each of P and R. A precipitate forms in each case. Describe a further chemical test that the student can complete on the precipitates to identify the halide ion present in P and in R. Describe how the observations can be used to identify each halide ion.

[3 marks]

Mark Scheme

M1, M2... are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

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Question	Marking guidance	Comments	Mark
01	$Ag^+(aq) + I^-(aq) \rightarrow AgI(s)$	*Source: Practice question (written for this set)*	1
02	**D**	*Source: 7404/1 Nov 2021 Q19*	1
03	**D**	*Source: 7405/3 Jun 2022 Q18*	1
04	Colourless (solution)	*Source: 7405/1 Nov 2020 Q09.1* Allow no (visible) change / no reaction / no ppt; ignore none or nothing	1
05	$Mg(NO_3)_2$: no visible change $Ba(NO_3)_2$: white precipitate	*Source: Practice question (written for this set)*	2
06	Add (dilute) sulfuric acid / any soluble sulfate $BaCl_2$ white precipitate; KCl no visible change	*Source: Practice question (written for this set)* Both observations needed for second mark	2
07.1	W NH_4Cl **and** Y $MgSO_4$ X Na_2CO_3 **and** Z KBr	*Source: Practice question (written for this set)* 2 correct = 1 mark	2
07.2	$NH_4^+ + OH^- \rightarrow NH_3 + H_2O$		1
07.3	$Mg^{2+} + 2OH^- \rightarrow Mg(OH)_2$		1
07.4	Add dilute ammonia — precipitate does not dissolve Add concentrated ammonia — precipitate dissolves	Allow: dissolves in conc NH_3 but not dilute	2
08.1	Iodide ion / I^- $Br_2 + 2I^- \rightarrow I_2 + 2Br^-$	*Source: 7404/1 Jun 2024 Q01.1, 01.3* Not iodine / I_2 Allow $Br_2 + 3I^- \rightarrow I_3^- + 2Br^-$	2
08.2	M1 add dilute ammonia M2 the precipitate containing chloride ions dissolves / forms a colourless solution M3 the precipitate containing bromide ions shows no visible change / does not dissolve	Or dilute then concentrated ammonia: bromide precipitate dissolves only in concentrated ammonia	3